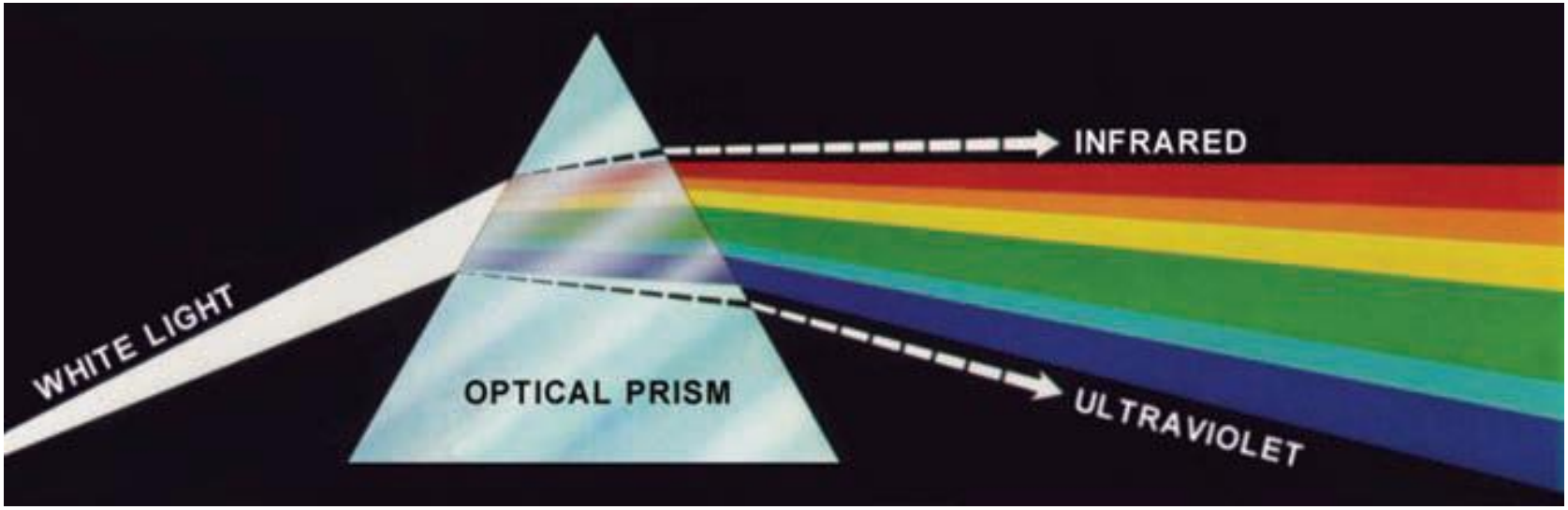
# Lecture 3.1

**Course Name: Digital Image Processing**
**Course code: 4245**

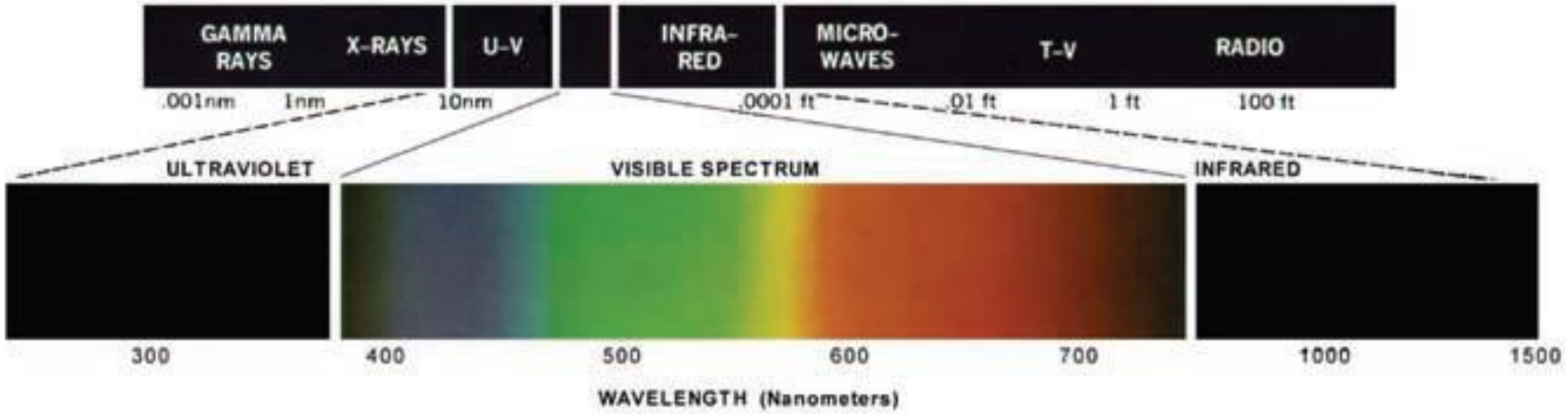
# Color Image Processing

Instructor Name:
**Professor Dr. Jahangir Alam**
**Professor, Dept. of CSE, BAUST**

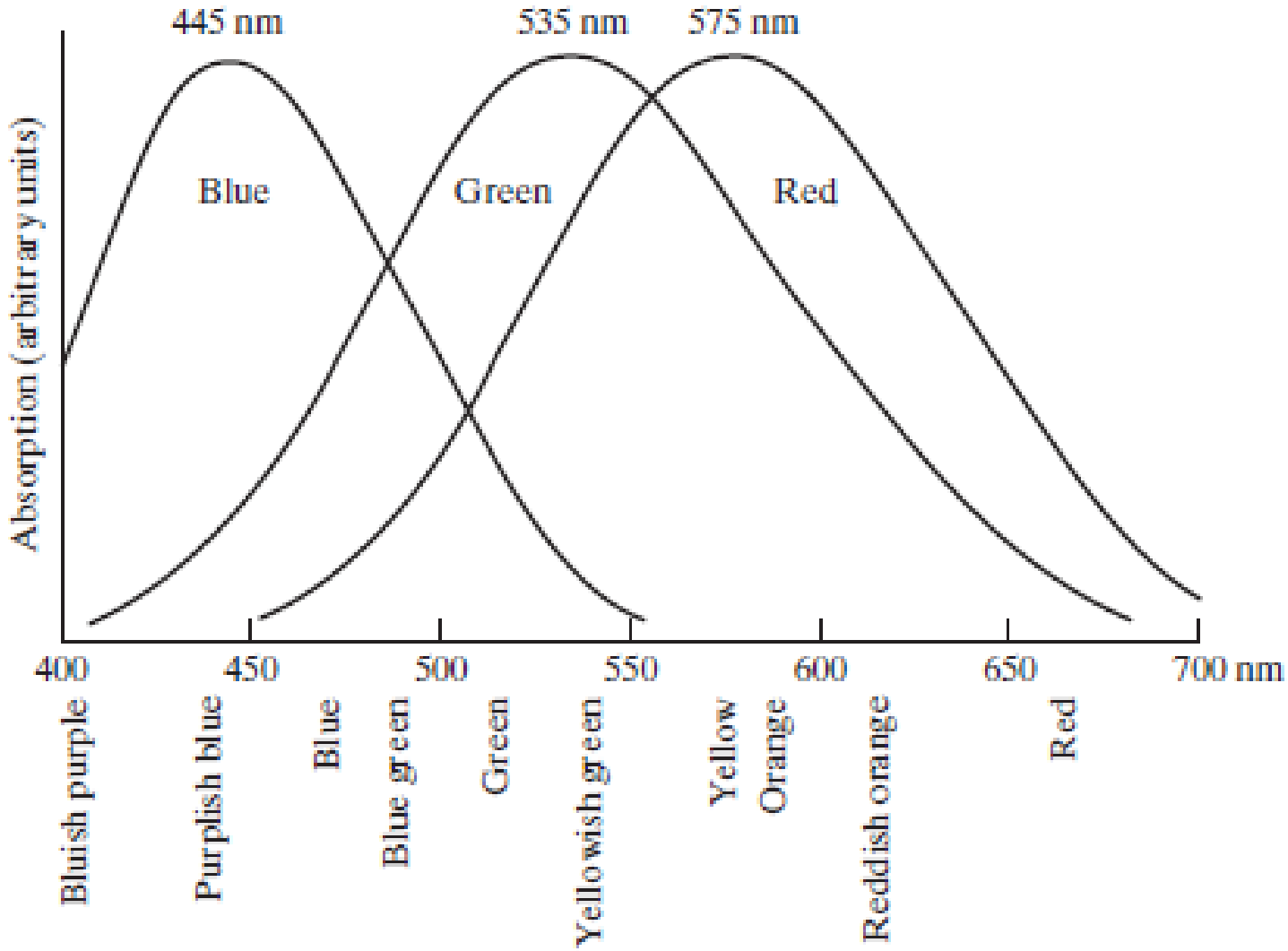
# Color Fundamentals



# Wavelengths comprising the visible range of the electromagnetic spectrum.

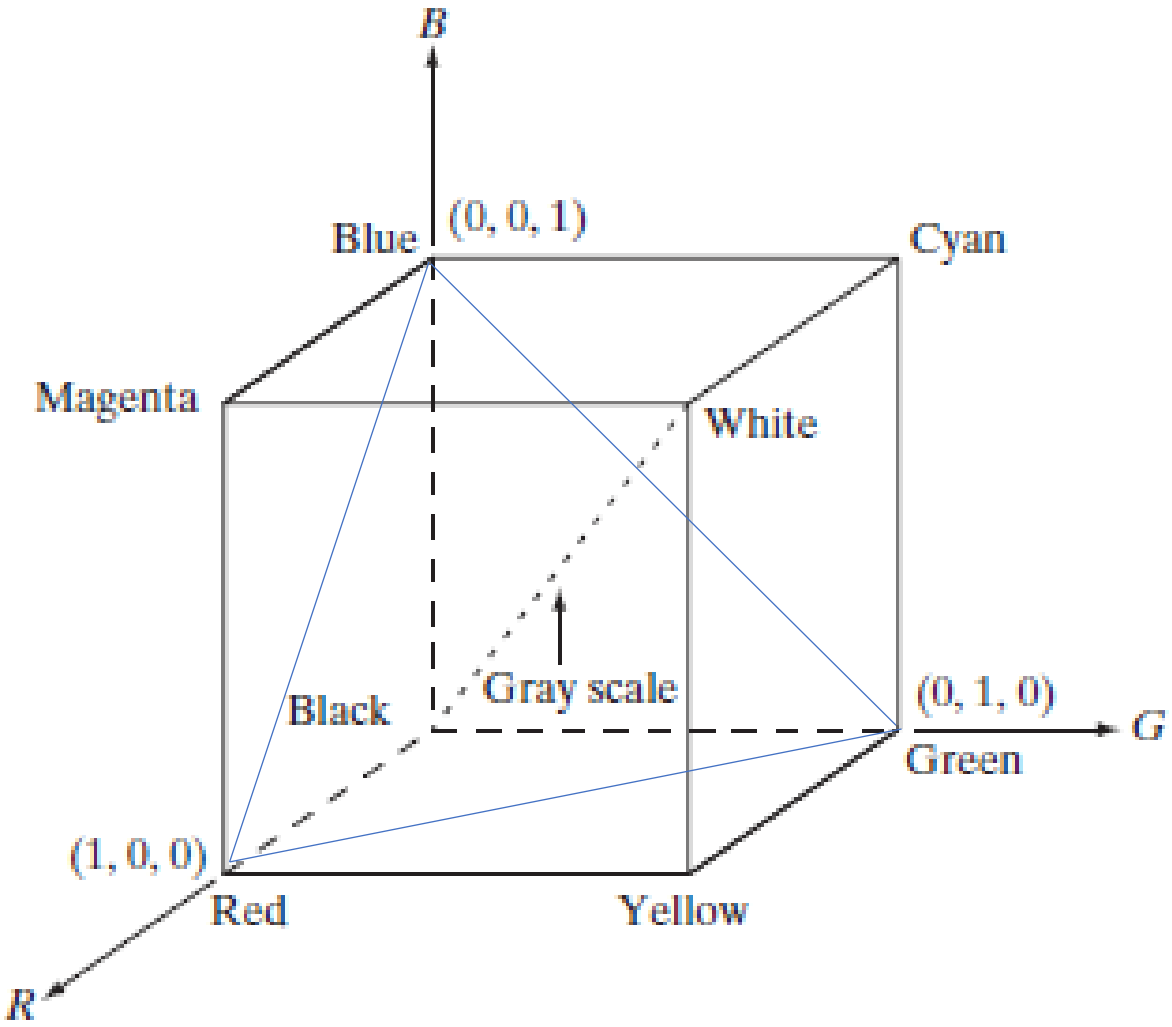


# Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.



# RGB Color Model

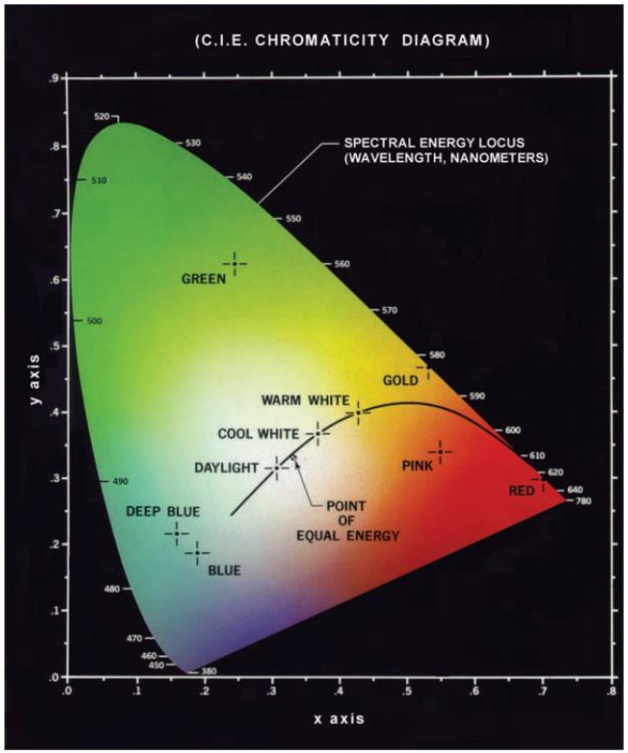
Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).

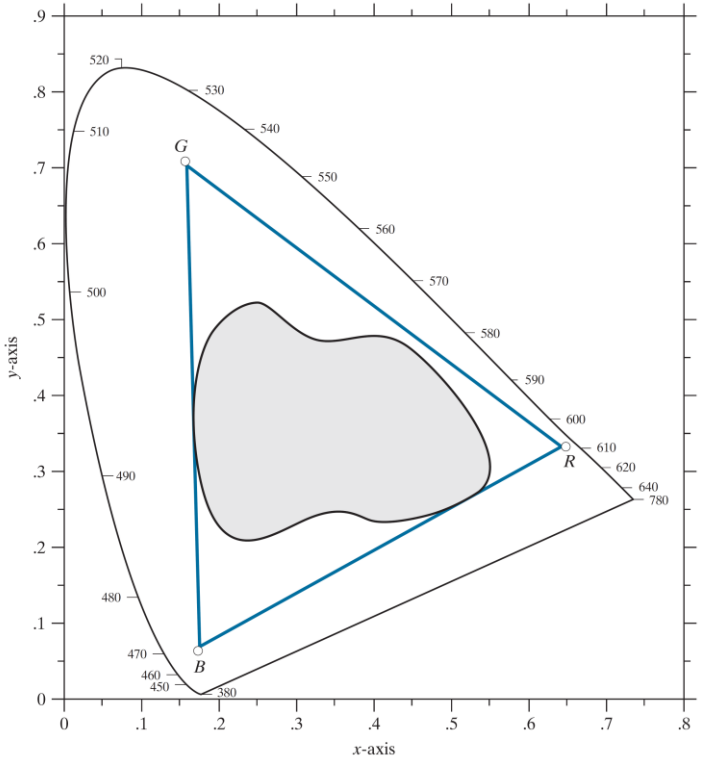


# RGB Color Model

$24-bit\ image$: $(2^8)^3$=16,777,216





# Color Image Processing

•Deals with **images represented in multiple color channels** (usually 3 for RGB).

•Each pixel is represented as a **vector of color values** rather than a single intensity value (as in grayscale).

•Goals: Enhance, analyze, or manipulate the **color information** for better visualization, segmentation, or recognition.

**Color image representation:**

$$\text{Pixel} = [R, G, B]^T \ (for\ RGB\ color\ model)$$

*Other color spaces often used:*

•**HSV / HSL**: Hue, Saturation, Value / Lightness-uses for segmentation.

•**YCbCr / YUV**: Luminance + Chrominance – widely used in video compression.

•**CIE XYZ / Lab**: Perceptually uniform color spaces.

# Color Models and Their Roles

**a) RGB (Red, Green, Blue)**

- Additive color model used in displays.
- Each channel: 0–255 (8-bit), combining to form color.
- **Processing challenge:** Correlation between channels, not perceptually uniform.

**b) HSV / HSL**

- Separates **color information** from **intensity/brightness**.
- Useful for color segmentation and enhancement.

**c) YCbCr / YUV**

- Luminance (Y) separated from chrominance (Cb, Cr).
- Common in **compression** (JPEG, MPEG).

**d) Lab / CIE XYZ**

- Lab: Luminance (L), two chromatic components (a, b).
- Perceptually uniform → useful in color difference measurement.

# Color Image Processing Techniques

**A) Color Transformation**

- Change color representation: RGB ↔ HSV, RGB ↔ YCbCr.
- **Example:** Converting to HSV to modify saturation or Hue.

**B) Color Enhancement (**Enhance contrast or brightness per channel)

- **Techniques:**
  - Histogram equalization on each channel.
  - Adaptive histogram equalization.
  - Color balancing (white balance correction).

**C) Color Segmentation (**Separate objects based on color).

- **Steps:**
  - Convert to perceptual color space (HSV, Lab).
  - Threshold Hue/Saturation values.
- Applications: Object detection, tracking, medical imaging.

# Color Image Processing Techniques

**D) Color Filtering and Smoothing:** Apply filters per channel,

- Gaussian smoothing, median filtering.
- Edge detection on luminance channel to avoid color artifacts.

**E) Color Edge Detection**

•Detect edges using **gradient magnitude** of each channel or **luminance only**.

•Avoids false edges due to color noise.

**F) Color Compression**

•Reduce storage by converting RGB → YCbCr and **subsampling chrominance** (e.g., 4:2:0).

•Used in JPEG, MPEG formats.

**G) Color Morphology / Object Analysis**

•Treat color images as **vector images**.

•Perform morphological operations like dilation, erosion in multi-channel space.

# Hue, Lab / CIE XYZ, YCbCr / YUV

- **Hue** is the **type of color** (RGB, etc.).

- Measured as an **angle (0°–360°: R=0°, G=120°, B=360°,…. etc.)** on a color wheel.

**HSV Components:**

- **H (Hue)** → color type
- **S (Saturation)** → purity of color
- **V (Value)** → brightness

**Why Hue is important:**

- Separates **color from brightness**
- Very useful for **color segmentation**

**International Commission on Illumination** (French acronym **CIE**: *Commission Internationale de l'Éclairage*): world's leading authority on light, illumination, color, & visual perception. Founded in 1913, international standards for colorimetry, lighting measurement, and photometry that underpin industries from manufacturing to environmental design.

***Example:*** Detecting a red object → threshold Hue values only.

# Hue, Lab / CIE XYZ, YCbCr / YUV

Lab / CIE XYZ Color Models

These are **perceptually meaningful and standardized color spaces**.

- **L*** → Lightness
- **a*** → Green ↔ Red
- **b*** → Blue ↔ Yellow

- **X** → Red-green mixture
- **Y** → **Luminance (brightness)**
- **Z** → Blue component

**CIE XYZ**

- A **device-independent color space** defined by the International Commission on Illumination

- Based on human visual perception

- All values are **non-negative**

- Used as a **reference model** for other color spaces

# Hue, Lab / CIE XYZ, YCbCr / YUV

**Lab (CIELAB)**

- Derived from **CIE XYZ**
- Designed to be **perceptually uniform**

**Why Lab is powerful:**

- Equal changes ≈ equal visual changes
- Ideal for:
  - Color difference measurement
  - Image enhancement
  - Color correction

**YCbCr / YUV Color Models**

✓ Used mainly in **video and image compression**.

✓ Separate **brightness from color.**

- **Y** → Luminance (brightness)
- **Cb** → Blue chrominance
- **Cr** → Red chrominance

# Hue, Lab / CIE XYZ, YCbCr / YUV

**Why useful:**

- Human eye is more sensitive to brightness than color
- Enables **compression (chroma sub-sampling)**

Used in:

- JPEG
- MPEG
- Digital TV

**YUV**

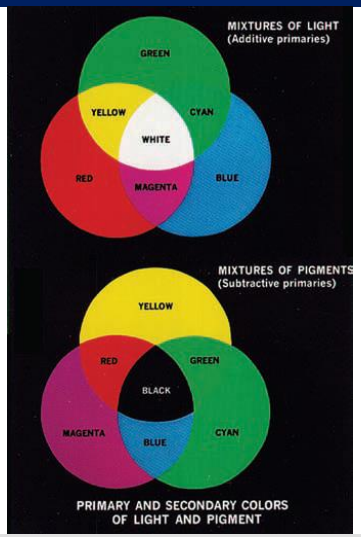
- Very similar to YCbCr
- Used in **analog video systems**

- **Y** → Luminance (brightness)
- **U** → Blue color difference
- **V** → Red color difference

Used mainly in **analog TV systems** and early video processing.

- **Hue (HSV)** → best for **detecting colors**
- **Lab / XYZ** → best for accurate color representation & analysis
- **YCbCr / YUV** → best for **compression and transmission**

# Primary and secondary colors of light and pigments.




**Colour combination:**

Green(255) + Red(255) = Yellow

Green(255) + Blue(255) = Cyan

Red(255) + Blue(255) = Magenta

Red(255) + Green(255) + Blue(255) = White

# CIE Chromaticity Diagram

**Equations:**

$$X = \int S(\lambda)\bar{x}(\lambda)\, d\lambda$$

$$Y = \int S(\lambda)\bar{y}(\lambda)\, d\lambda$$

$$Z = \int S(\lambda)\bar{z}(\lambda)\, d\lambda$$

$$r = \int S(\lambda) \cdot \bar{r}(\lambda)\, d\lambda$$

$$g = \int S(\lambda) \cdot \bar{g}(\lambda)\, d\lambda$$

$$b = \int S(\lambda) \cdot \bar{b}(\lambda)\, d\lambda$$

Where $\bar{r}(\lambda), \bar{g}(\lambda), \bar{b}(\lambda)$ are the **color matching functions** for the **chosen primary lights**.

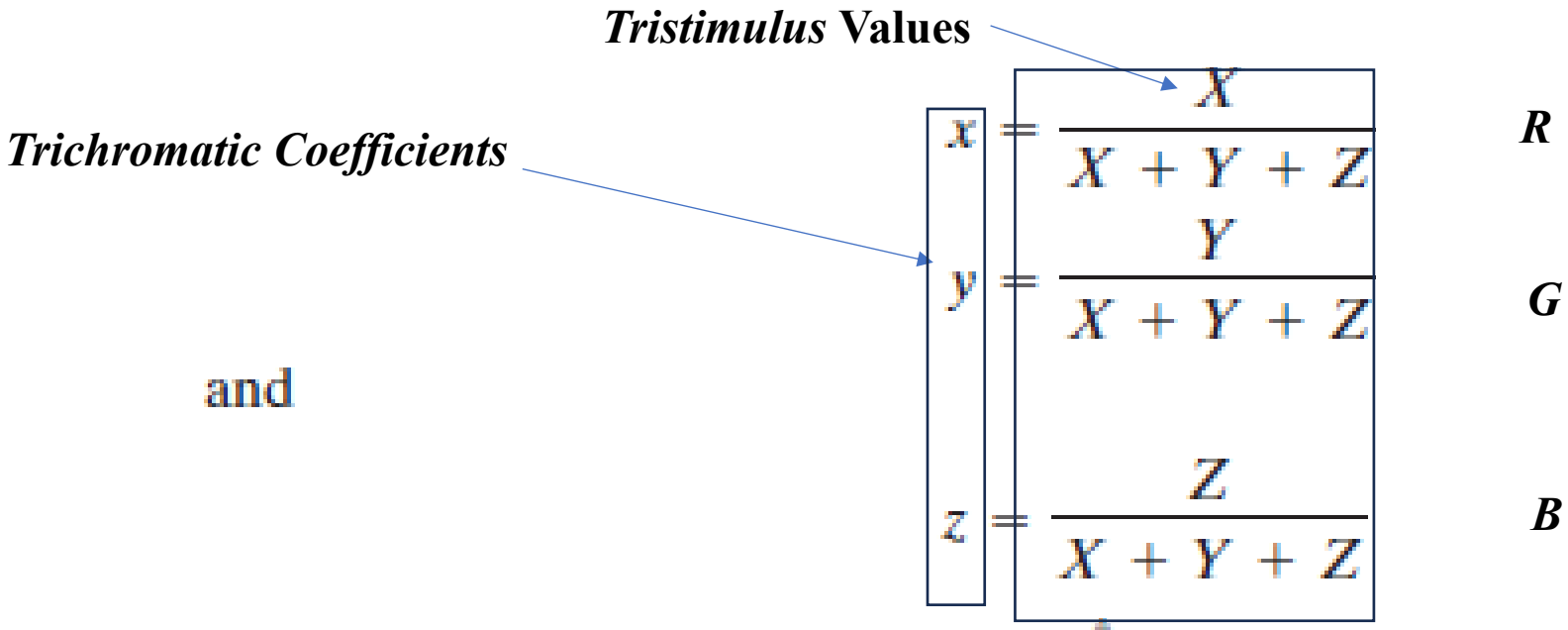
**Light spectrum , $S(\lambda)$:**

- Spectral power distribution of light
- Represents **intensity at each wavelength λ.**

**$d\lambda$ :**

- Small wavelength interval
- Integration sums contributions over all visible wavelengths

# CIE Chromaticity Diagram




$$r + g + b = 1 \text{ or, } \quad x + y + z = 1$$

**Trichromatic coefficients** (x,y,z)=$(r, g, b)$ represent **how much of each primary light (R, G, B) is needed to match the color of** $S(\lambda)$in a **color matching experiment**.

# RGB → HSV, YCbCr, and Lab (via XYZ)

Suppose we have only 3 wavelengths: Assume $\Delta\lambda=1$

| $\lambda$ | $S(\lambda)$ | $\bar{x}(\lambda)$ | $\bar{y}(\lambda)$ | $\bar{z}(\lambda)$ |
|---|---|---|---|---|
| $\lambda_1$ | 10 | 0.5 | 0.4 | 0.1 |
| $\lambda_2$ | 20 | 0.3 | 0.5 | 0.2 |
| $\lambda_3$ | 30 | 0.2 | 0.1 | 0.7 |

**Compute X**

$X=10(0.5)+20(0.3)+30(0.2) =17$

**Compute Y**

$Y=10(0.4)+20(0.5)+30(0.1) =17$

**Compute Z**

$Z=10(0.1)+20(0.2)+30(0.7) =26$

**Final Result:**
(X,Y,Z)=(17, 17, 26)

# *Trichromatic Coefficients vs Tristimulus* Values

**Example:**

- Let's say you have tristimulus values: X = 100, Y = 50, and Z = 25.
- Then, the trichromatic coefficients would be:
  - x = 100 / (100 + 50 + 25) = 100 / 175 = 0.5714
  - y = 50 / (100 + 50 + 25) = 50 / 175 = 0.2857
  - z = 25 / (100 + 50 + 25) = 25 / 175 = 0.1429
- Note that x + y + z = 0.5714 + 0.2857 + 0.1429 = 1.

# RGB → HSV, YCbCr, and Lab (via XYZ)

**Given RGB Pixel**

Let's take:

$$R = 100, G = 150, B = 200$$

Normalize (divide by 255):

$$R' = 0.392, G' = 0.588, B' = 0.784$$

**1. RGB → HSV**

**Step 1: Find max, min**

$$C_{max} = 0.784, C_{min} = 0.392$$

$$\Delta = C_{max} - C_{min} = 0.392$$

**Step 2: Hue (H)**

Since, $C_{max} = B$:

$$H = 60 \times \left(\frac{R' - G'}{\Delta} + 4\right)$$

$$= 60 \times \left(\frac{0.392 - 0.588}{0.392} + 4\right)$$

$$= 210°$$

**Step 3: Saturation (S)**

$$S = \frac{\Delta}{C_{max}} = \frac{0.392}{0.784} = 0.5$$

**Step 4: Value (V)**

$$V = C_{max} = 0.784$$

**HSV Result:**

$$(H, S, V) = (210°, 0.5, 0.784)$$

# RGB → HSV, YCbCr, and Lab (via XYZ)

**Given RGB Pixel**

Let's take:

$$R = 100, G = 150, B = 200$$

Normalize (divide by 255):

$$R' = 0.392, G' = 0.588, B' = 0.784$$

**YCbCr Result:**

$$(Y, Cb, Cr) \approx (\mathbf{140.75}, \mathbf{161.44}, \mathbf{98.94})$$

**XYZ Result:**

$$(X, Y, Z) \approx (\mathbf{0.512}, \mathbf{0.559}, \mathbf{0.821})$$

**RGB → YCbCr**

$$Y=0.299R+0.587G+0.114B = 140.75$$

$$Cb=128-0.168736R-0.331264G+0.5B= 161.44$$

$$Cr = 128 + 0.5R - 0.418688G - 0.081312B = 98.94$$

**RGB → CIE XYZ**

$$X = 0.412R' + 0.358G' + 0.180B' = 0.512$$
$$Y = 0.213R' + 0.715G' + 0.072B' = 0.559$$
$$Z = 0.019R' + 0.119G' + 0.950B' = 0.821$$

# RGB → HSV, YCbCr, and Lab (via XYZ)

**XYZ → Lab**

Use reference white (approx):

$$X_n = 0.950,$$
$$Y_n = 1.000,$$
$$Z_n = 1.089$$

**Step 1: Normalize**

$$\frac{X}{X_n} = 0.539, \frac{Y}{Y_n} = 0.559, \frac{Z}{Z_n} = 0.754$$

**Step 2: Cube root**

$$f(X) = 0.813, f(Y) = 0.824, f(Z) = 0.912$$

**Step 3: Compute Lab**

$$L = 116f(Y) - 16 = 116(0.824) - 16 = 79.6$$

$$a = 500\big(f(X) - f(Y)\big) = 500(0.813 - 0.824) = -5.5$$

$$b = 200\big(f(Y) - f(Z)\big) = 200(0.824 - 0.912) = -17.6$$

**(L,a,b)≈(79.6, –5.5, –17.6)**

# RGB → HSV, YCbCr, and Lab (via XYZ)

**Given RGB Pixel**

Let's take:

$$R = 100, G = 150, B = 200$$

**RGB → YUV Formula**

For RGB values in range **0–255**:

$$Y = 0.299R + 0.587G + 0.114B$$
$$U = -0.147R - 0.289G + 0.436B$$
$$V = 0.615R - 0.515G - 0.100B$$

**Step 1: Compute Y**

$$Y = 0.299(100) + 0.587(150) + 0.114(200) = 140.75$$

**Step 2: Compute U**

$$U = -0.147(100) - 0.289(150) + 0.436(200) = 29.15$$

**Step 3: Compute V**

$$V = 0.615(100) - 0.515(150) - 0.100(200) = -35.75$$

**Final YUV Result:**

$$(Y, U, V) = (140.75, 29.15, -35.75)$$

# RGB → HSV, YCbCr, and Lab (via XYZ)

**Given RGB Pixel**

Let's take:

$R = 100, G = 150, B = 200$

**YUV → RGB (Inverse)**

$$R = Y + 1.14V$$
$$G = Y - 0.395U - 0.581V$$
$$B = Y + 2.032U$$

Used to reconstruct original image.

**Step 1: Compute R**

$$R = Y + 1.14 \cdot V = 140.75 + 1.14 \cdot (-35.75) \approx 100$$

**Step 2: Compute G**

$$G = Y - 0.395 \cdot U - 0.581 \cdot V$$
$$= 140.75 - 0.395 \cdot 29.15 - 0.581 \cdot (-35.75) \approx 150.0$$

**Step 3: Compute B**

$$B = Y + 2.032 \cdot U = 140.75 + 2.032 \cdot 29.15 \approx 200$$

**Resulting RGB:**

$$(R, G, B) \approx (100,150,200)$$

# CMY & CMYK COLOR MODELS

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

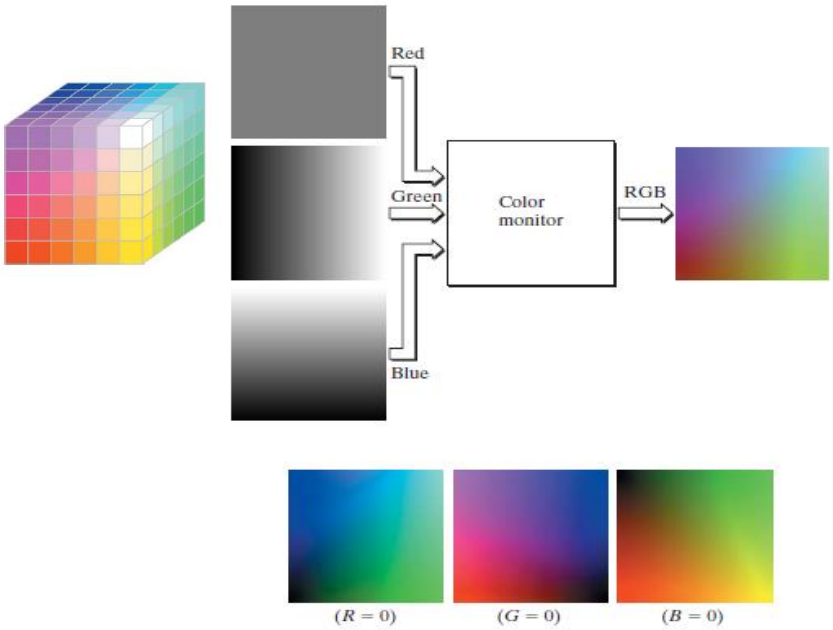
$$K = \min(C, M, Y)$$

$$C = 0$$

$$M = 0$$

$$Y = 0$$

- **C** = Cyan
- **M** = Magenta
- **Y** = Yellow




The **CMY color model** is a **subtractive color model**, meaning colors are created by subtracting light.

- Colors are produced by **absorbing (subtracting)** certain wavelengths
  - Cyan absorbs Red
  - Magenta absorbs Green
  - Yellow absorbs Blue

**Key Idea:**

**C + M + Y** ⇒ muddy dark (not pure black).

# CMY & CMYK COLOR MODELS

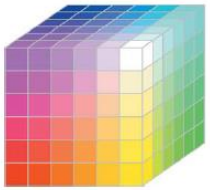
$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

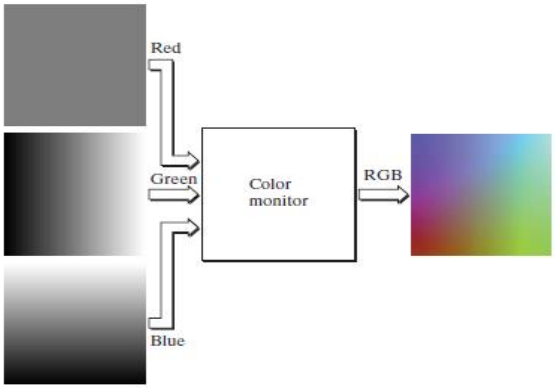
$$K = \min(C, M, Y)$$

$$C = 0$$

$$M = 0$$

$$Y = 0$$

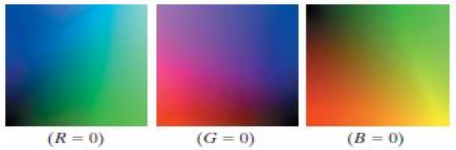





**CMYK Color Model**

The **CMYK color model** improves CMY by adding black.

**Components: K** = Black (Key).




**Why add Black (K)?**

- CMY alone cannot produce deep black
- Adding **K**:
  - Improves contrast
  - Saves ink
  - Produces sharper text and images

**Relation with RGB**

- CMY/CMYK is used in **printing**
- RGB is used in **displays**
- They are **inverse systems**:
  - $C = 255 - R$
  - $M = 255 - G$
  - $Y = 255 - B$

# CMY & CMYK COLOR MODELS

**CMY Calculation:**

- **Cyan (C):**

$$C = \begin{cases} 1 - \frac{R}{255} \\ 1 - R \end{cases}$$ ;If RGB is normalized to 0-1

- **Magenta (M):**

$$M = \begin{cases} 1 - \frac{G}{255} \\ 1 - G \end{cases}$$ ;If RGB is normalized to 0-1

- **Yellow (Y):**

$$Y = \begin{cases} 1 - \frac{B}{255} \\ 1 - B \end{cases}$$ ;If RGB is normalized to 0-1

# CMY & CMYK COLOR MODELS

RGB value from 1 (or 255, depending on the RGB representation). Convert RGB(200, 100, 50) to CMY.

Sol. Calculate, C=1-R=0.20, M=1-G=0.50, and Y=1-B=0.50, resulting in CMY (0.20, 0.50, 0.50)

Let's say you have the RGB color (200, 100, 50).

C = 1 - (200 / 255) ≈ 0.22, M = 1 - (100 / 255) ≈ 0.61, and Y = 1 - (50 / 255) ≈ 0.80.

Therefore, the CMY equivalent of (200, 100, 50) is approximately (0.22, 0.61, 0.80).

# CMY & CMYK COLOR MODELS

$$C = \frac{C' - K}{1 - K}$$

$$C = C' \times (1 - K) + K$$

$$M = \frac{M' - K}{1 - K}$$

$$M = M' \times (1 - K) + K$$

$$Y = \frac{Y' - K}{1 - K}$$

$$Y = Y' \times (1 - K) + K$$

Let's say we have a CMY color value of (0.6, 0.3, 0.9).

$$K = \min(C, M, Y)$$

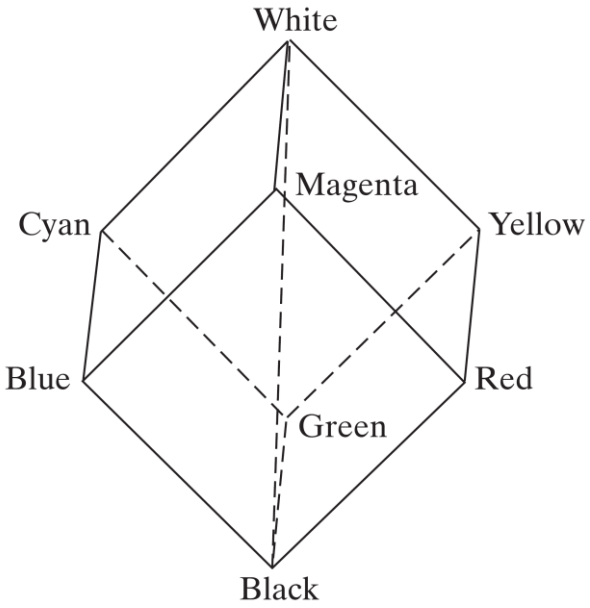
K = 0.3

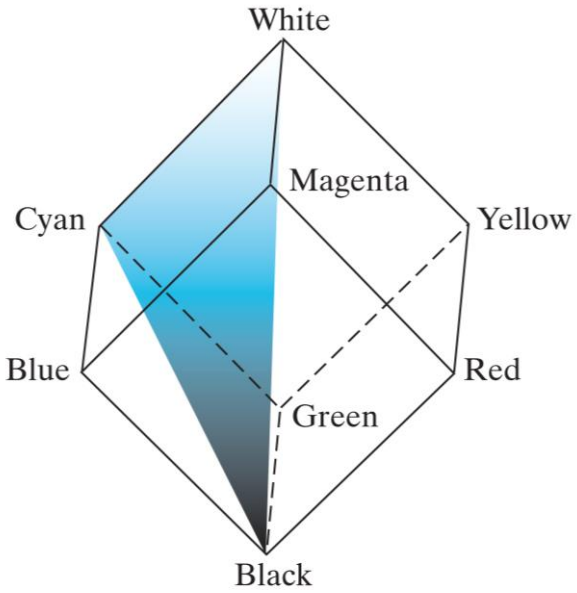
$$C = \frac{0.6 - 0.3}{1 - 0.3} = 0.43$$

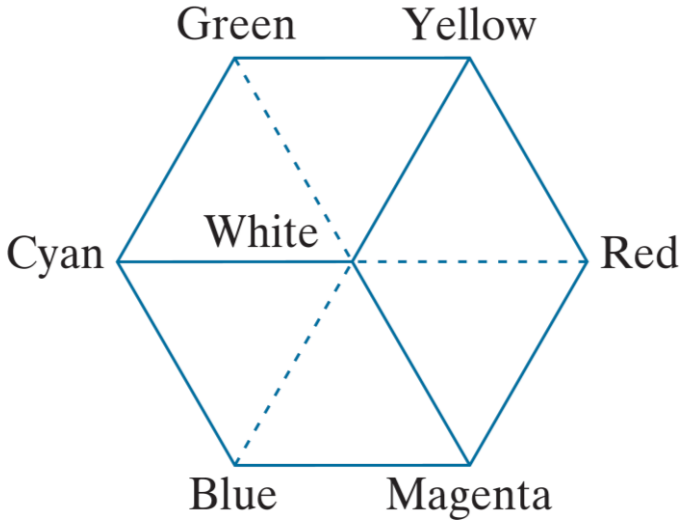
$$M = \frac{0.3 - 0.3}{1 - 0.3} = 0.0$$

$$Y = \frac{0.9 - 0.3}{1 - 0.3} = 0.86$$

# HSI Color Model

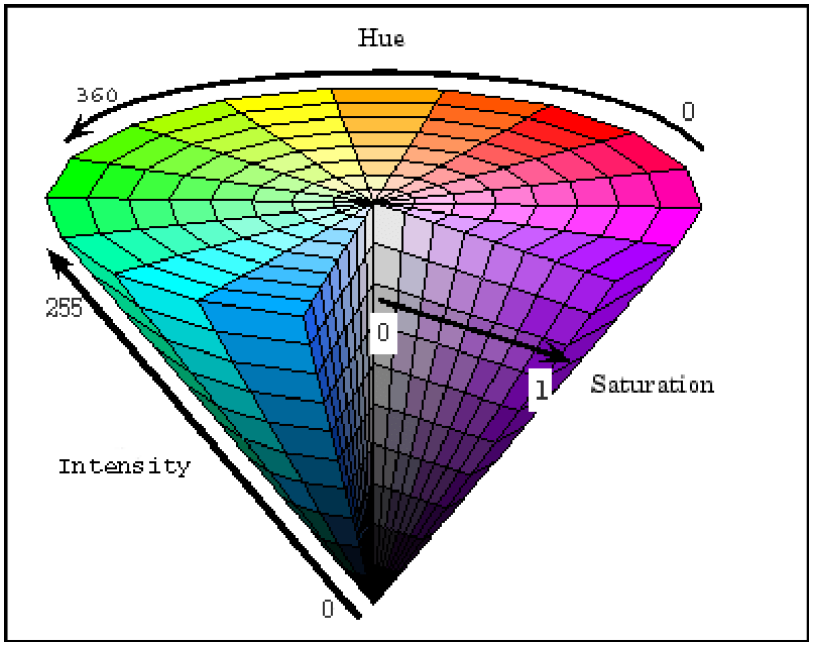


# HSI Color Model

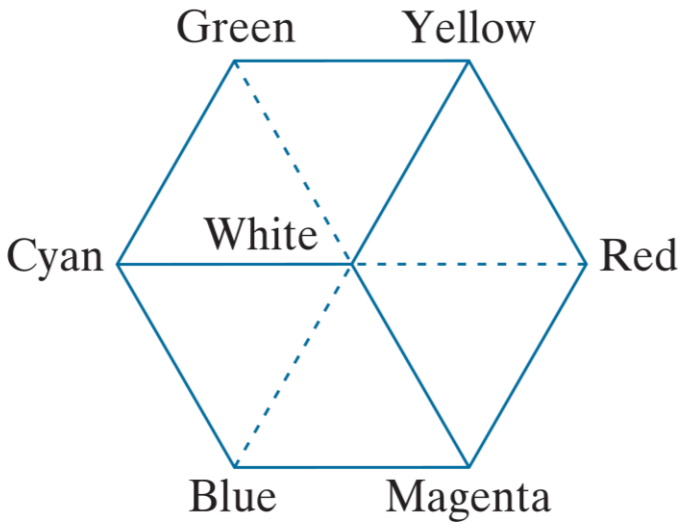
**Hue (H):** Represents the color itself, typically described as an angle between 0 and 360 degrees. For example, 0 degrees is red, 120 degrees is green, and 240 degrees is blue.

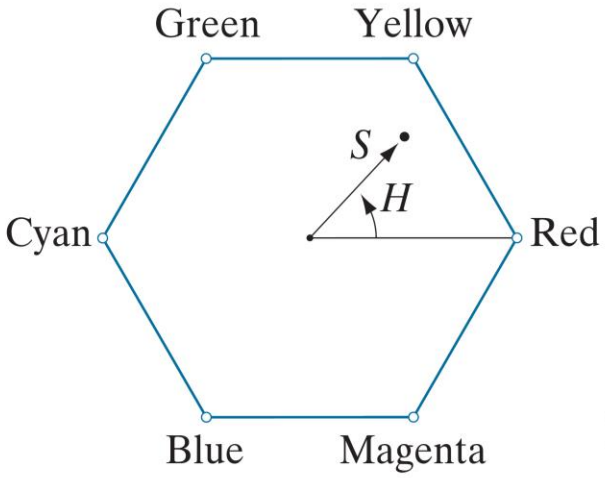


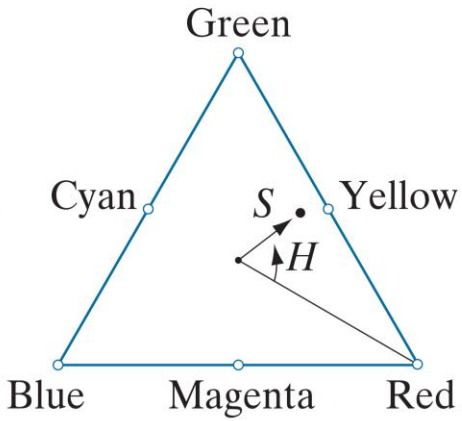

**Saturation (S):** Represents the purity or intensity of the color. A fully saturated color is pure, while a less saturated color is diluted with white. Saturation values range from 0 (gray) to 1 (fully saturated).

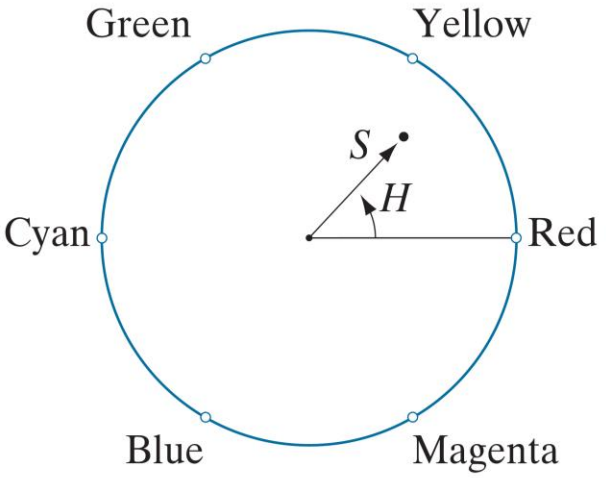
**Intensity (I):** Represents the brightness or lightness of the color, ranging from 0 (black) to 1 (white).

# HSI Color Model

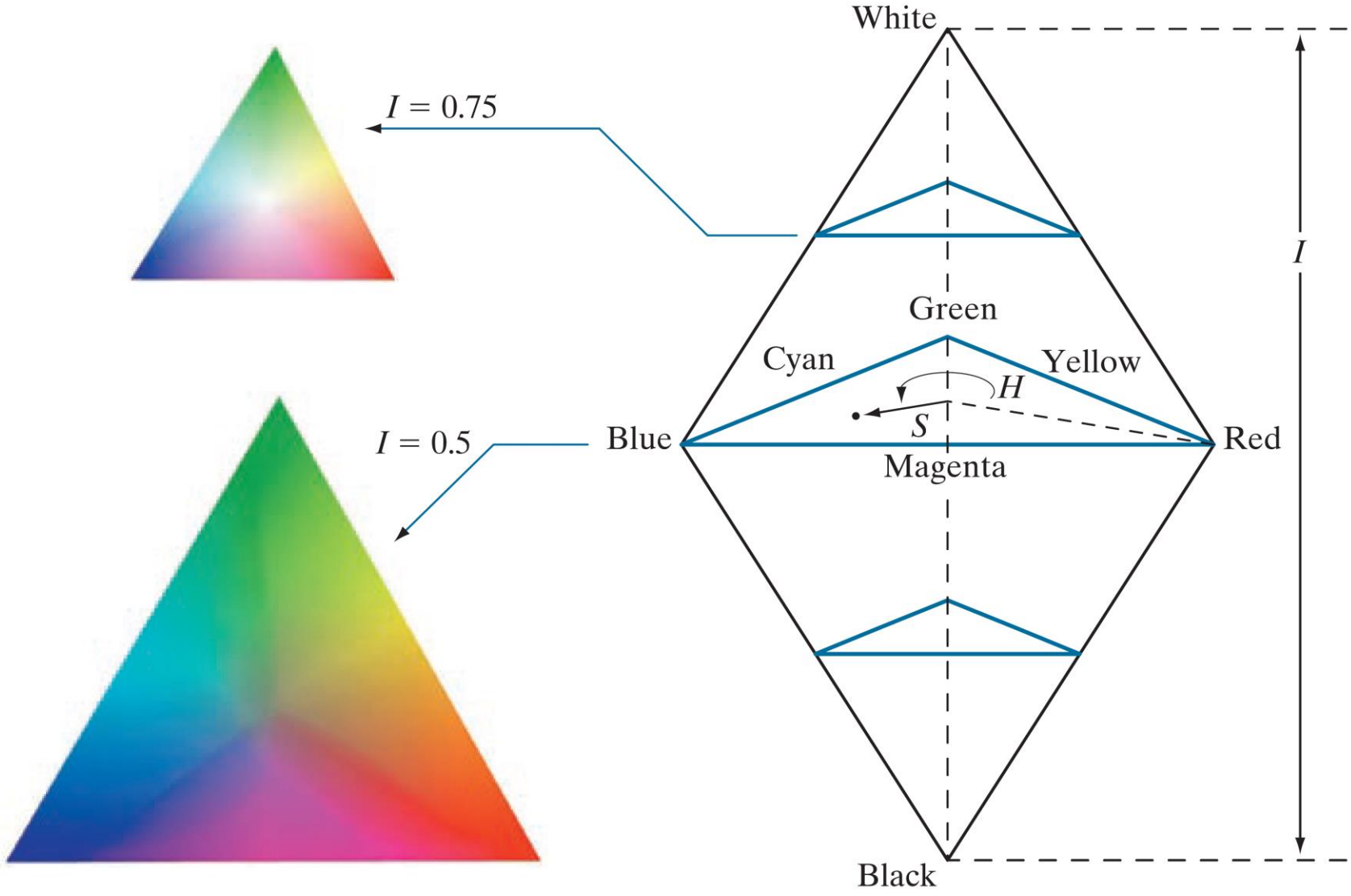


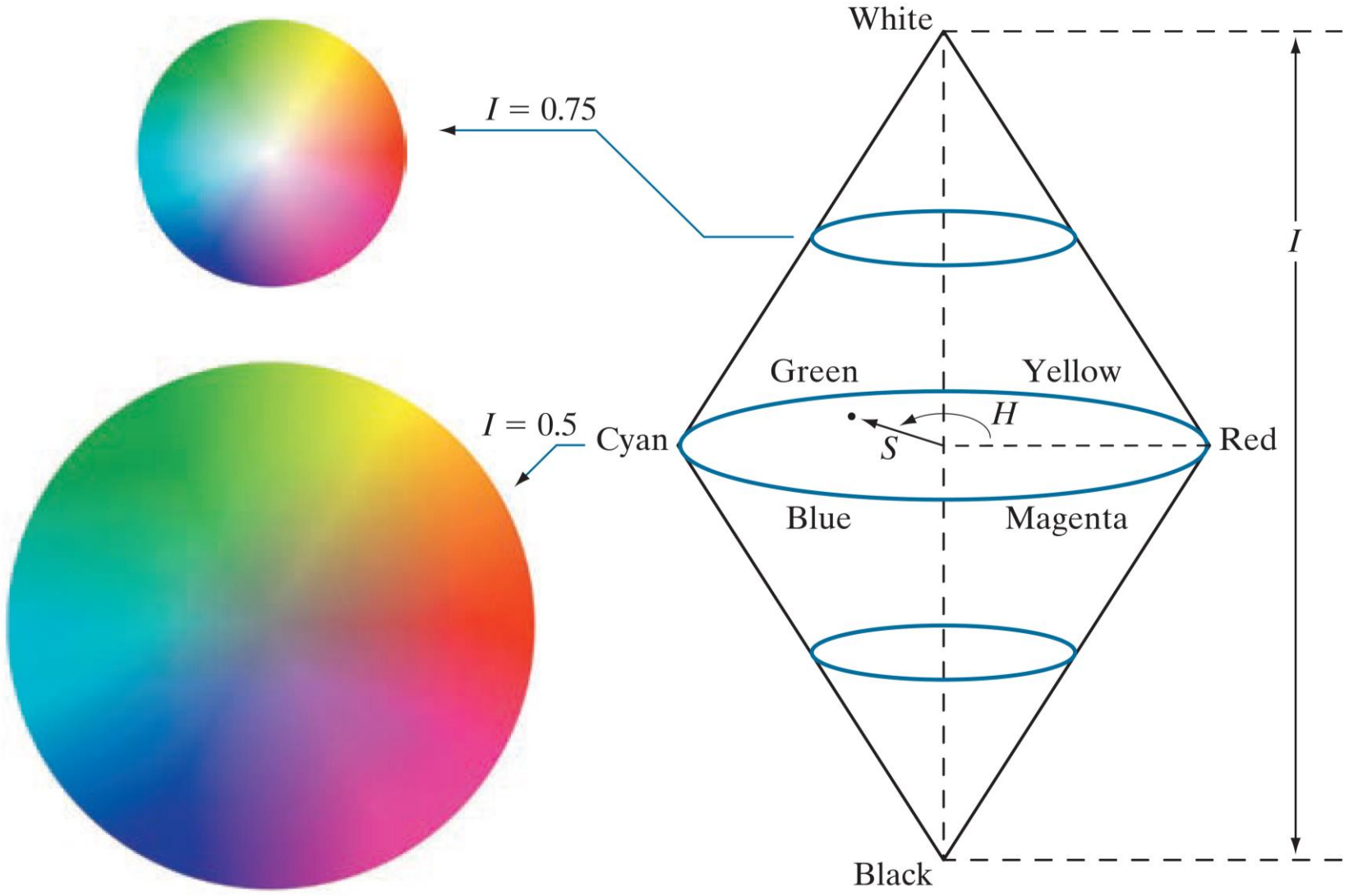



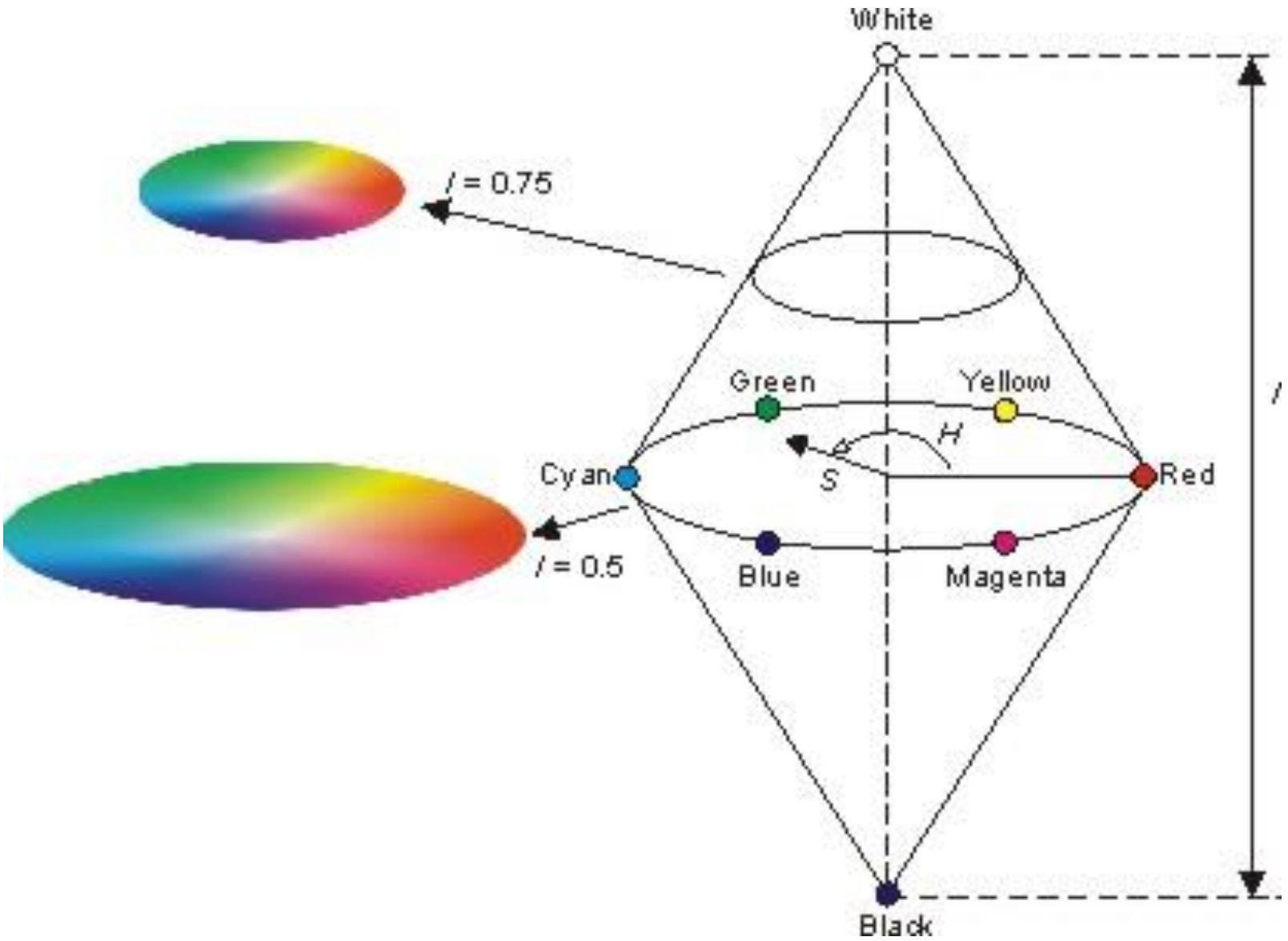
# HSI Color Model



# HSI Color Model



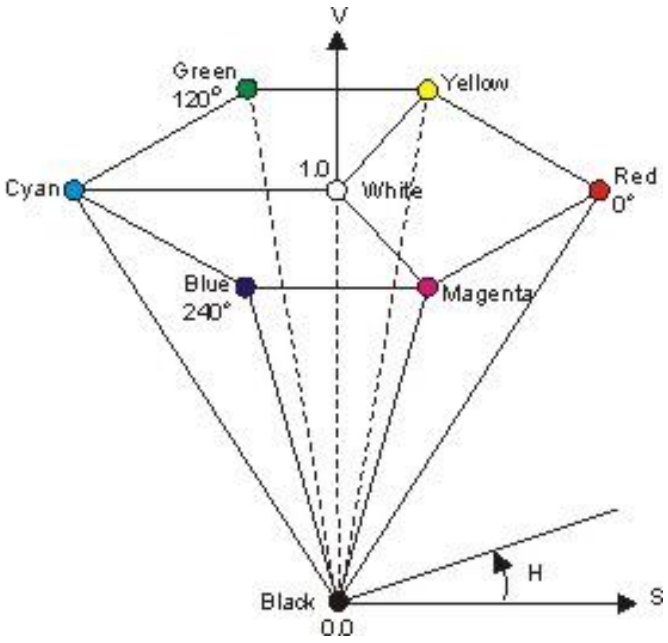
# HSI Color Model



# HSI Color Model

To formula that converts from RGB to HSI or back is more complicated than with other color models, therefore we will not elaborate on the detailed specifics involved in this process.

| | | |
|---|---|---|
| RGB | CMY | CMYK |
| HSI | HSV | L*a*b |
| XYZ | YIQ | YUV |




The HSV color space is very similar to the HSI color space. In HSV every color is represented by three components Hue ( H ), Saturation ( S ) and Value ( V ).

# Converting Colors from RGB to HSI

$$H = \begin{cases} \theta & \text{if } B \le G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$\theta = \cos^{-1}\left\{\frac{\frac{1}{2}\left[(R-G)+(R-B)\right]}{\left[(R-G)^2+(R-B)(G-B)\right]^{1/2}}\right\}$$

$$S = 1 - \frac{3}{(R+G+B)}\left[\min(R,G,B)\right]$$

$$I = \frac{1}{3}(R+G+B)$$

# Converting Colors from HSI to RGB

***RG sector*** $(0° \le H < 120°)$:

$$B = I(1 - S)$$

$$R = I\left[1 + \frac{S\cos H}{\cos(60° - H)}\right]$$

$$G = 3I - (R + B)$$

# Converting Colors from HSI to RGB

***GB sector*** $(120° \le H < 240°)$

$$H = H - 120°$$

$$R = I(1 - S)$$

$$G = I\left[1 + \frac{S\cos H}{\cos(60° - H)}\right]$$

$$B = 3I - (R + G)$$

# Converting Colors from HSI to RGB

***BR sector*** $(240° \le H \le 360°)$

$$H = H - 240°$$

$$G = I(1 - S)$$

$$B = I\left[1 + \frac{S\cos H}{\cos(60° - H)}\right]$$

$$R = 3I - (G + B)$$

# Converting Colors from RGB to HSI

Example:

Let's say we have an RGB color value of (100, 200, 50).

**1.Normalize:**

R = 100/350 = 0.392,

G = 200/350 = 0.784,

B = 50/350 = 0.196

$$\theta = \cos^{-1}\left\{\frac{\frac{1}{2}\left[(R-G)+(R-B)\right]}{\left[(R-G)^2+(R-B)(G-B)\right]^{1/2}}\right\}$$

**2.Calculate H:** H ≈ 100°

**3.Calculate S:** S ≈ 0.96

**4.Calculate I:** I ≈ 0.505

$$S = 1 - \frac{3}{(R+G+B)}\left[\min(R,G,B)\right] \qquad I = \frac{1}{3}(R+G+B)$$

# Converting Colors from HSI to RGB

Example:

Let's say you have an HSI color with H = 120, S = 0.5, and I = 200.

***GB sector*** $(120° \le H < 240°)$

$$H = H - 120° \qquad R = I(1 - S)$$

$$G = I\left[1 + \frac{S\cos H}{\cos(60° - H)}\right] \qquad B = 3I - (R + G)$$

# Converting Colors from HSI to RGB

Example:

Let's say you have H = 30°, S = 0.8, and I = 0.5.

***RG sector*** $(0° \le H < 120°)$:

$$B = I(1 - S)$$

$$R = I\left[1 + \frac{S\cos H}{\cos(60° - H)}\right]$$

$$G = 3I - (R + B)$$

# Converting Colors from HSI to RGB

1. Input HSI Values:

Let's assume we have an HSI color with the following values: H (Hue) = 60 degrees, S (Saturation) = 0.7, and I (Intensity) = 0.8.

***RG sector*** $(0° \le H < 120°)$:

$$B = I(1 - S)$$

$$R = I\left[1 + \frac{S\cos H}{\cos(60° - H)}\right]$$

$$G = 3I - (R + B)$$

Result:

The RGB values for the given HSI color are: R = 0.8, G = 0.8, and B = 0.24.

# Lecture 3.2

**Course Name: Digital Image Processing**
**Course code: 4245**

# Color Image Processing

Instructor Name:
**Professor Dr. Jahangir Alam**
**Professor, Dept. of CSE, BAUST**

# L*a*b Color Components

$$L^* = 116 \cdot h\left(\frac{Y}{Y_W}\right) - 16$$

Lightness

$$a^* = 500\left[h\left(\frac{X}{X_W}\right) - h\left(\frac{Y}{Y_W}\right)\right]$$

Red minus green

$$b^* = 200\left[h\left(\frac{Y}{Y_W}\right) - h\left(\frac{Z}{Z_W}\right)\right]$$

Green minus blue

$$h(q) = \begin{cases} \sqrt[3]{q} & q > 0.008856 \\ 7.787q + 16/116 & q \le 0.008856 \end{cases}$$

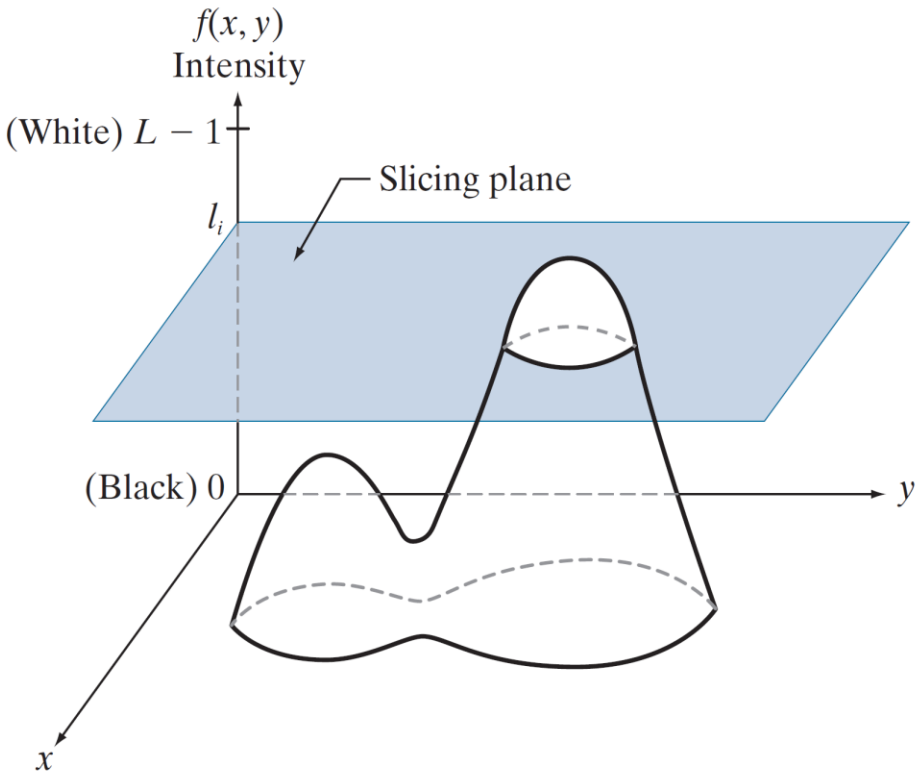
# Intensity (Density) Slicing & Color Coding

$f(x, y) = l_i$ to slice the image function into two levels.

$l_0(Black)$: $f(x, y) = 0$

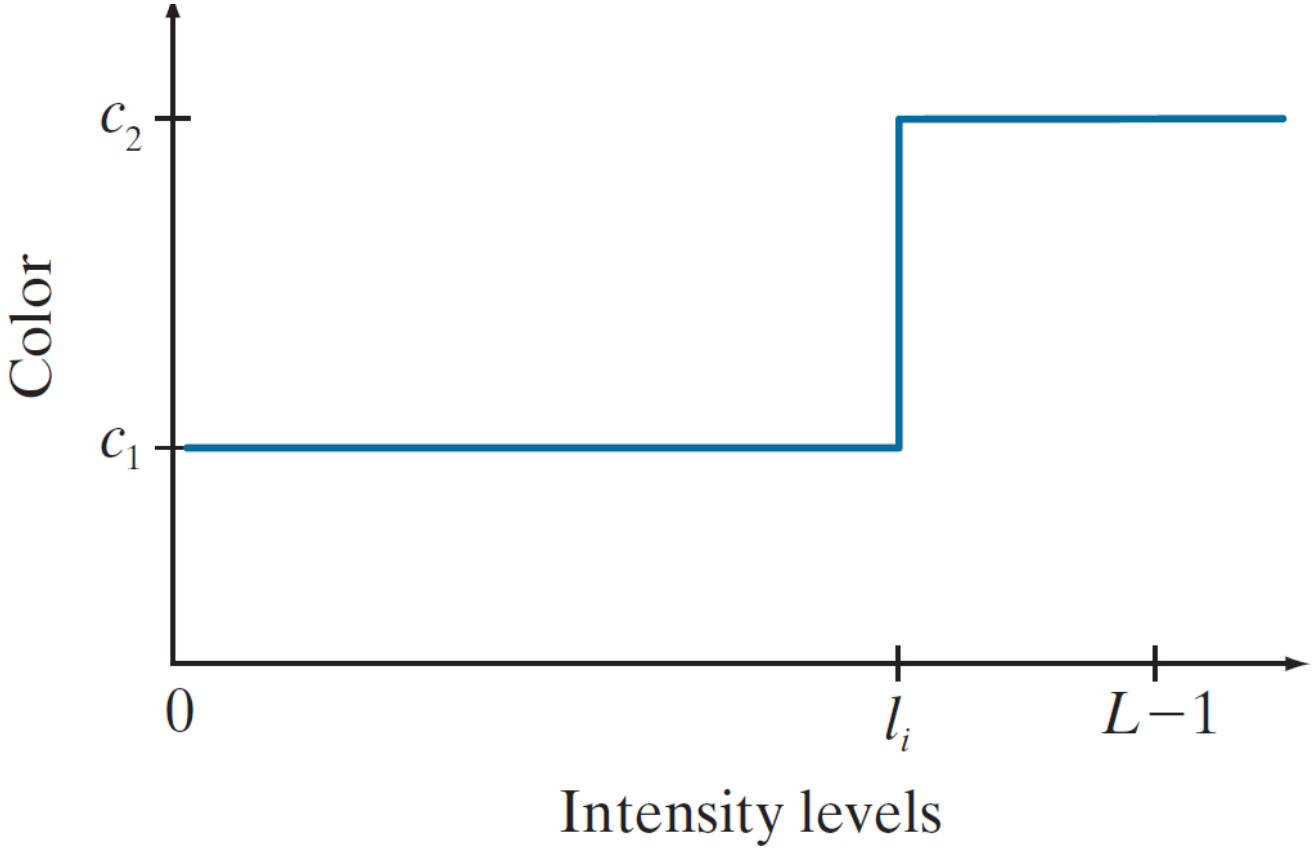
$l_{L-1}(White)$: $f(x, y) = L - 1$

if $f(x, y) \in I_k$, let $f(x, y) = c_k$



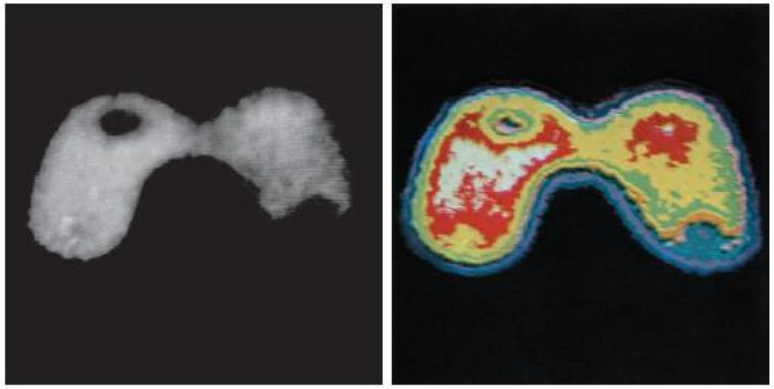
# Intensity (Density) Slicing & Color Coding

if $f(x,y) \in I_k$, let $f(x,y) = c_k$



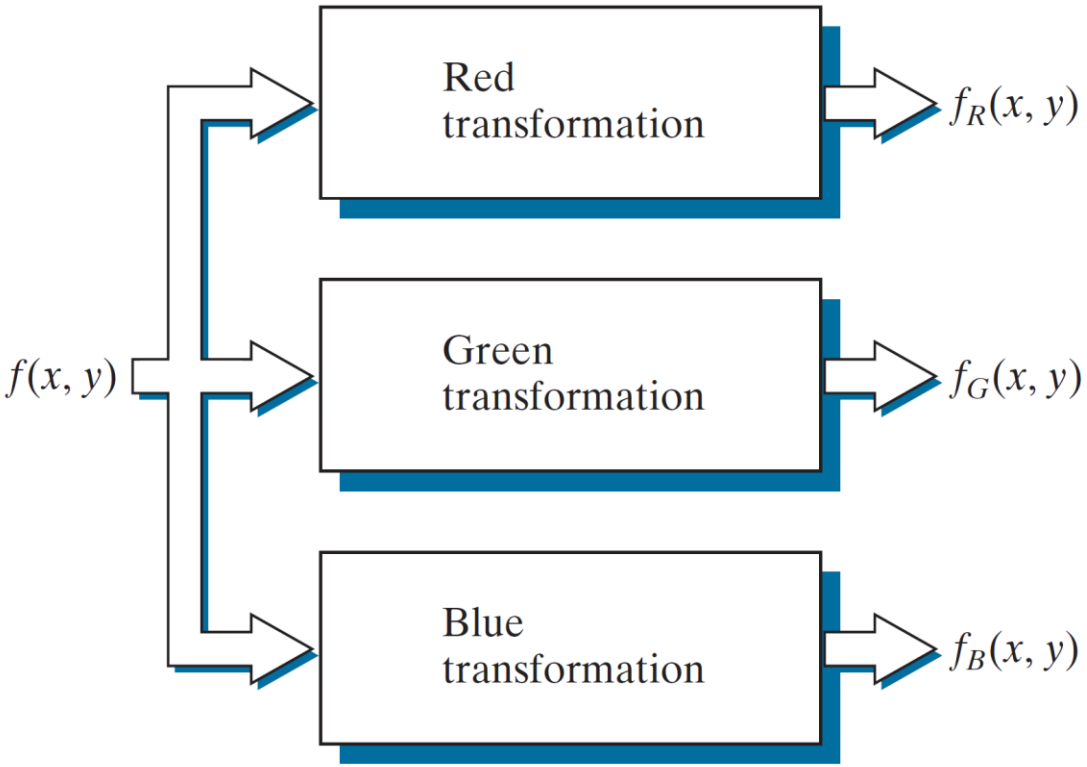
# Example: Picker Thyroid Phantom (radiation test pattern)

- Result of intensity slicing this image into 8 color regions.
- Regions that appear of constant intensity in the monochrome image are really quite variable,
- various colors in the sliced image.
- The left lobe is a dull gray in the monochrome image, and
- picking out variations in intensity is difficult.
- By contrast, the color image clearly shows 8 different regions of constant intensity, one for each of the colors used.



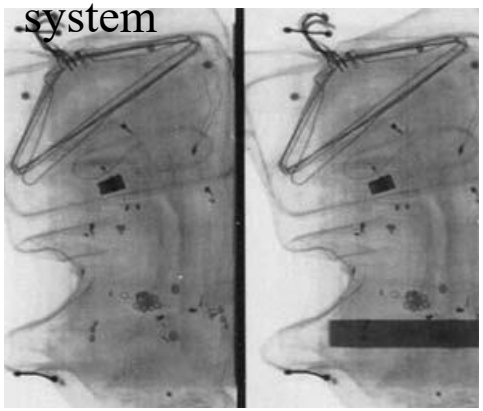
monochrome image

# Intensity to Color Transformations



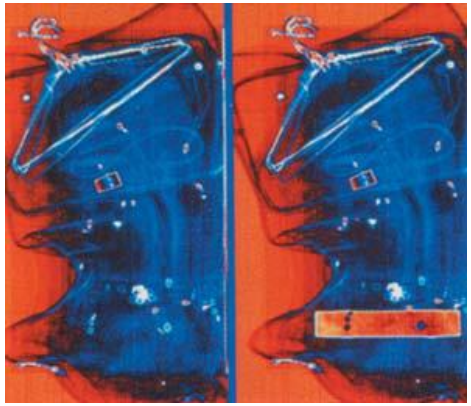
# Use of pseudocolor for highlighting explosives contained in luggage

Two monochrome images of luggage obtained from an airport X-ray scanning system

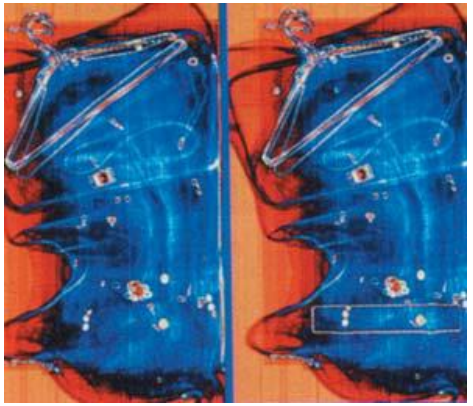


Using Transformation Function

contains ordinary articles

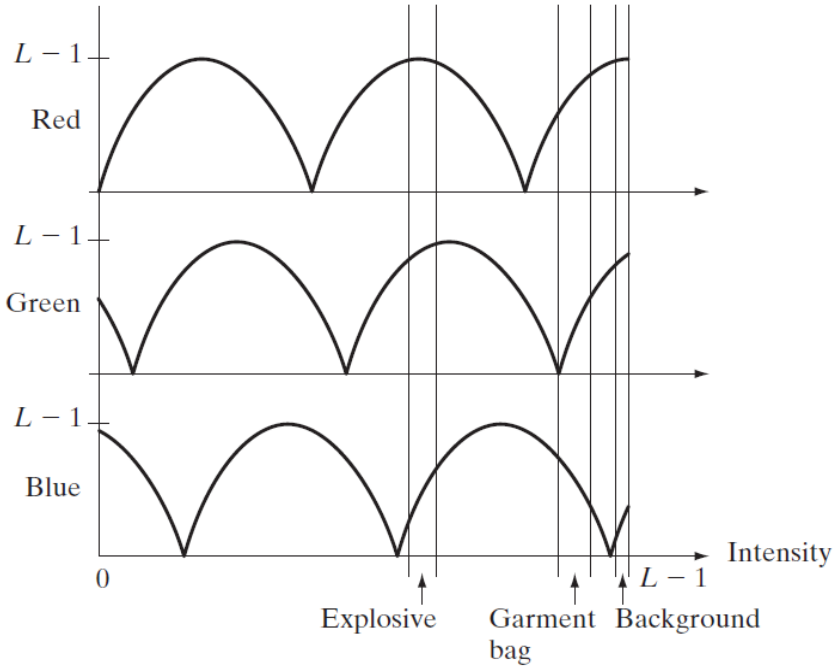


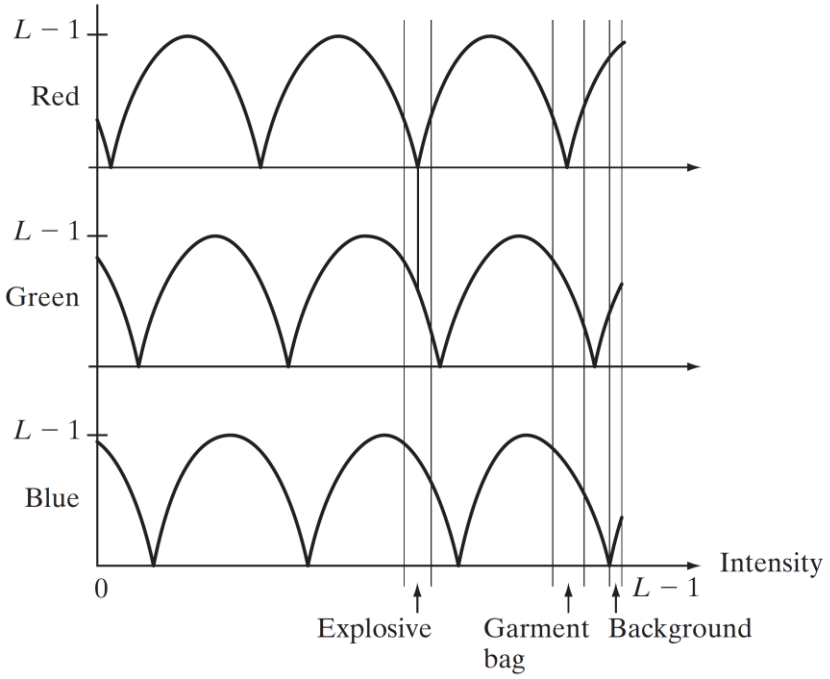
Plastic explosive



**Explosive, garment, background**

# Use of Pseudocolor for Highlighting Explosives Contained in Luggage



# Basics of Full-Color Image Processing

$$\mathbf{c} = \begin{bmatrix} c_R \\ c_G \\ c_B \end{bmatrix} = \begin{bmatrix} R \\ G \\ B \end{bmatrix} \Rightarrow \mathbf{c}(x,y) = \begin{bmatrix} c_R(x,y) \\ c_G(x,y) \\ c_B(x,y) \end{bmatrix} = \begin{bmatrix} R(x,y) \\ G(x,y) \\ B(x,y) \end{bmatrix}$$

$x = 0,1,2,..,M-1$

$y = 0,1,2,..,N-1$

# Color Transformations

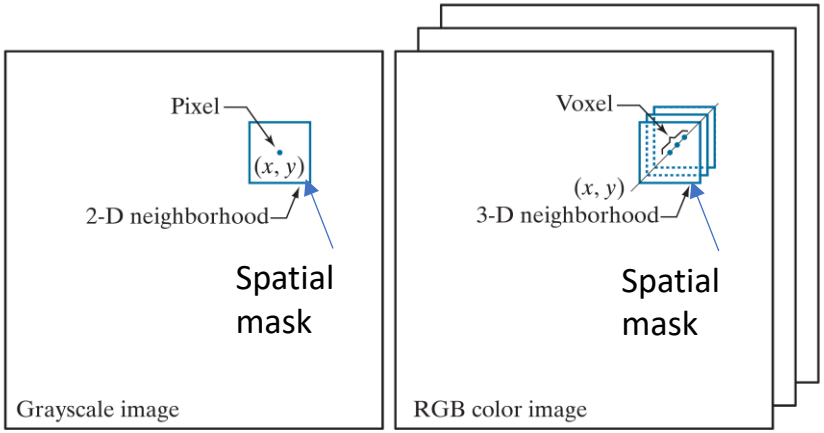
RGB-to-HSI and HSI-to-RGB conversion transformations:

$$g(x,y) = T[f(x,y)]$$

$$s_i = T_i(r_i) \qquad i = 1,2,\ldots,n$$

set of *transformation* or *color mapping functions*

number of color components



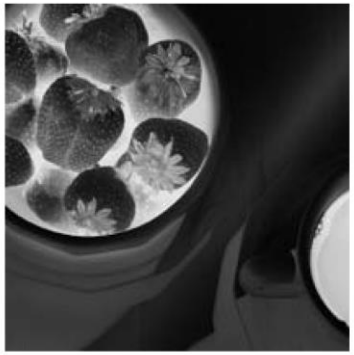

$n = 3$, RGB color space is selected,
$r_1 = Red, r_2 = Green, r_3 = Blue$

$n = 4$ *or* $3$, CMYK/HSI color space,
$r_1 = C, r_2 = M, r_3 = Y, r_4 = K$

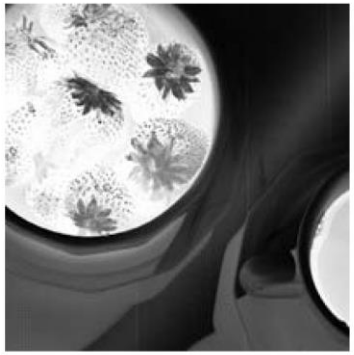
# Color Transformations



Full color



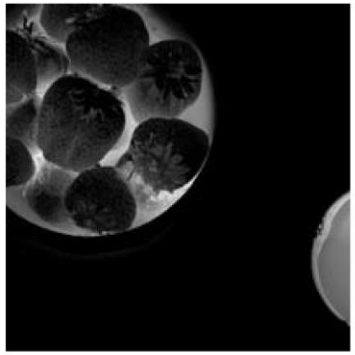
Cyan



Magenta



Yellow



Black



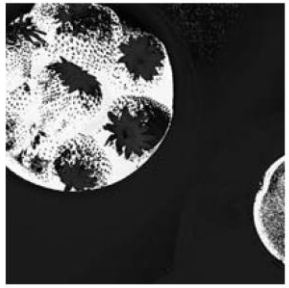
Red



Green



Blue



Hue



Saturation



Intensity

# Color Transformations

RGB-to-HSI and HSI-to-RGB conversion transformations:

$$g(x,y) = kf(x,y) \qquad 0 < k < 1$$

HIS color space:

$$s_3 = kr_3 \qquad s_1 = r_1 \;\&\; s_2 = r_2, for\ HSI, r_3 \text{ is modified}$$

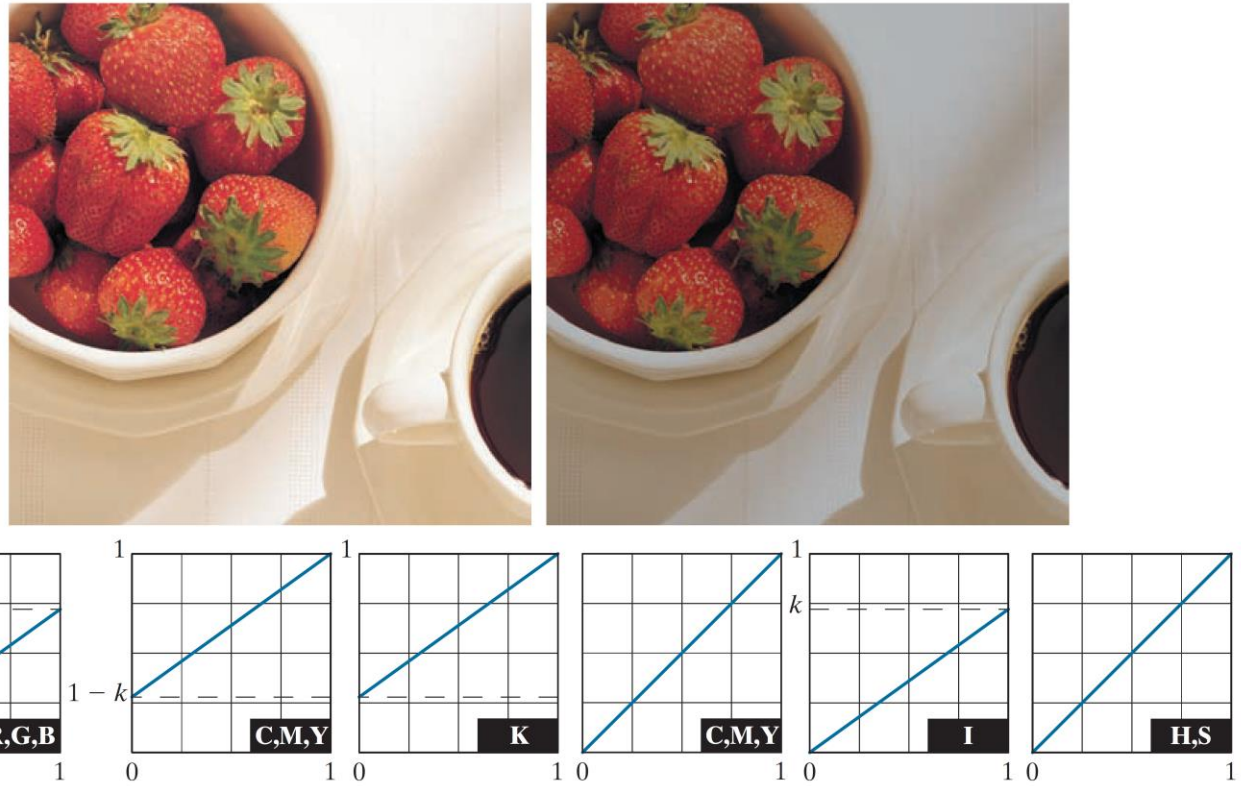
RGB color space:

$$s_i = kr_i \qquad i = 1,2,3$$

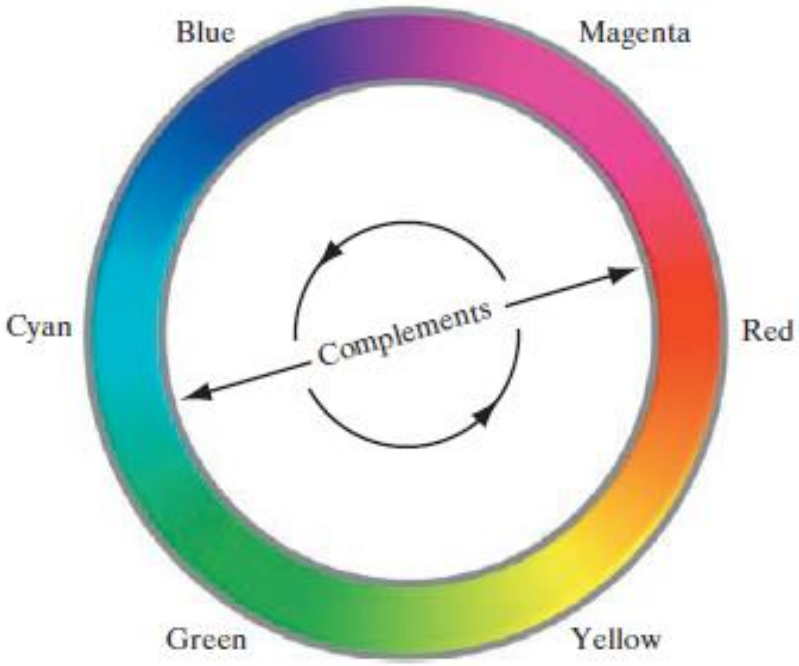
CMYK color space:

$$s_i = kr_i + (1-k) \qquad i = 1,2,3 \Rightarrow s_i = \begin{cases} r_i & i = 1,2,3 \\ kr_i + (1-k) & i = 4 \end{cases}$$

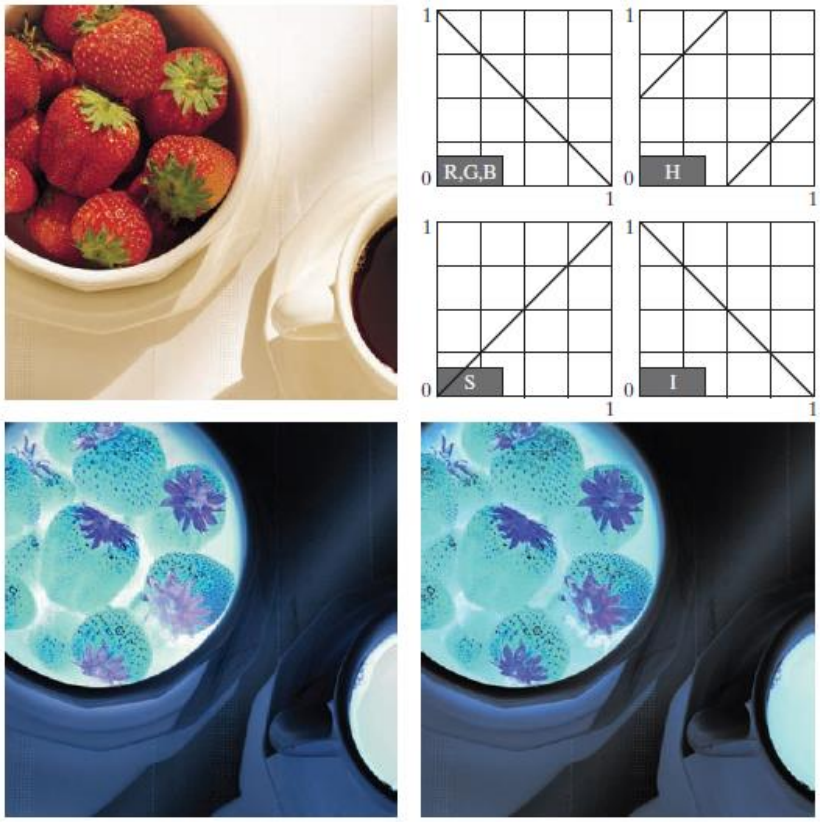
# Color Transformations



# Color Complements



# Color Complements



# Color Slicing

$$s_i = \begin{cases} 0.5 & \text{if } \left[ |r_j - a_j| > \frac{W}{2} \right]_{\text{any } 1 \le j \le n} \\ r_i & \text{otherwise} \end{cases}$$

Hypercube *for* $n>3$

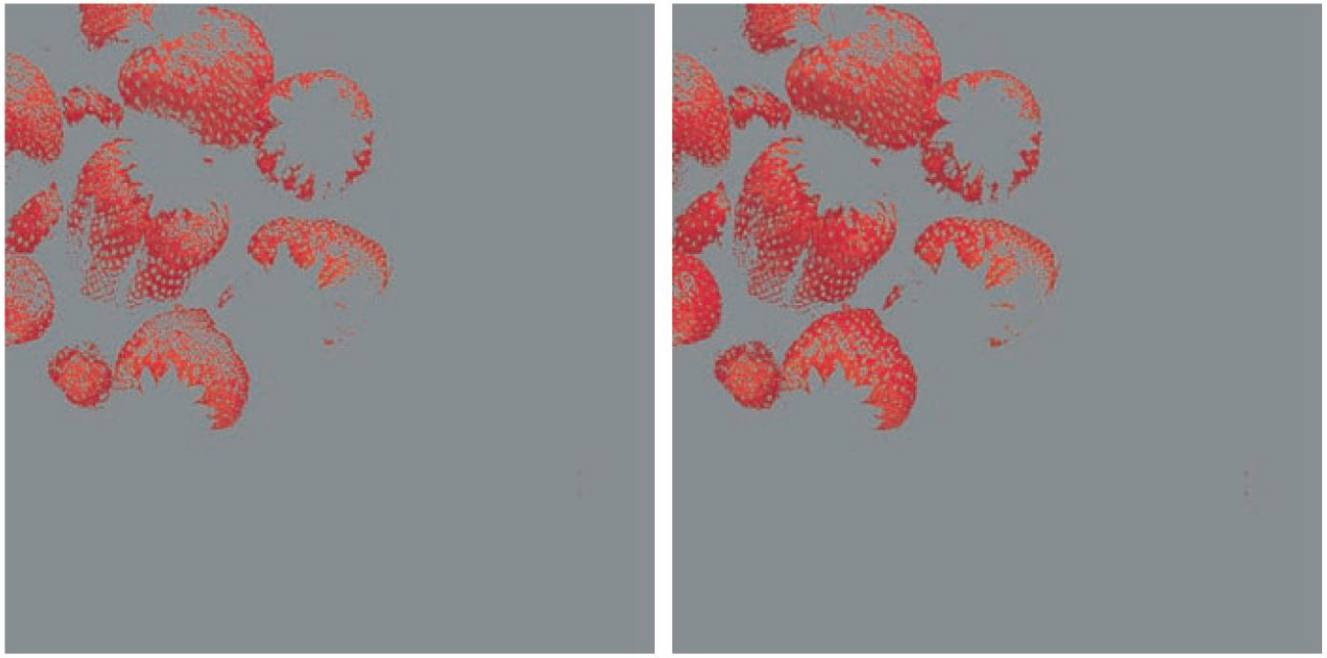
$$i = 1, 2, \ldots, n$$

Hypersphere *for* $n>3$

$$s_i = \begin{cases} 0.5 & \text{if } \sum_{j=1}^{n} (r_j - a_j)^2 > R_0^2 \\ r_i & \text{otherwise} \end{cases}$$

$$i = 1, 2, \ldots, n$$

# Color Slicing



a b

**FIGURE 6.32** Color-slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at $(0.6863, 0.1608, 0.1922)$, and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color $(0.5, 0.5, 0.5)$.

# Color Slicing

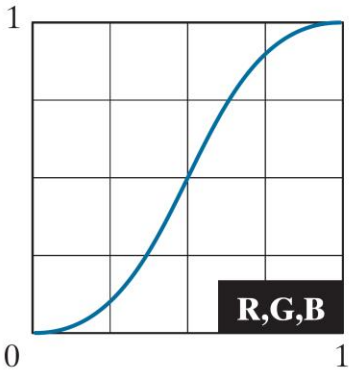
**Example:** Transformations for modifying image tones normally are selected interactively. The idea is to adjust experimentally the image's brightness and contrast to provide maximum detail over a suitable range of intensities. The colors themselves are not changed. In the RGB and CMY(K) spaces, this means mapping all three (or four) color components with the same transformation function; in the HSI color space, only the intensity component is modified.



Flat



Corrected

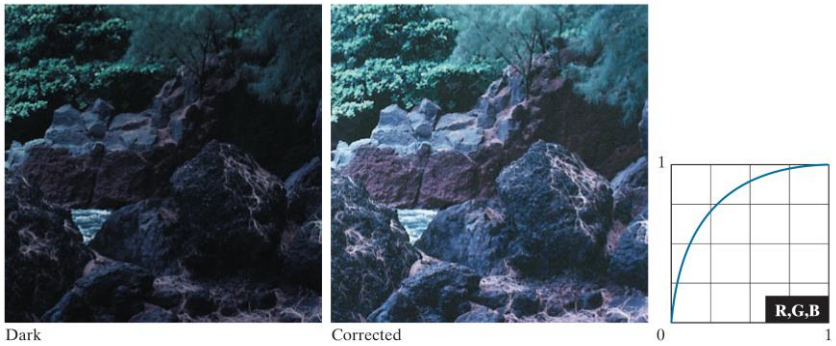



Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting RGB components equally does not always alter the image hues significantly.

# Color Corrections

Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting RGB components equally does not always alter the image hues significantly.

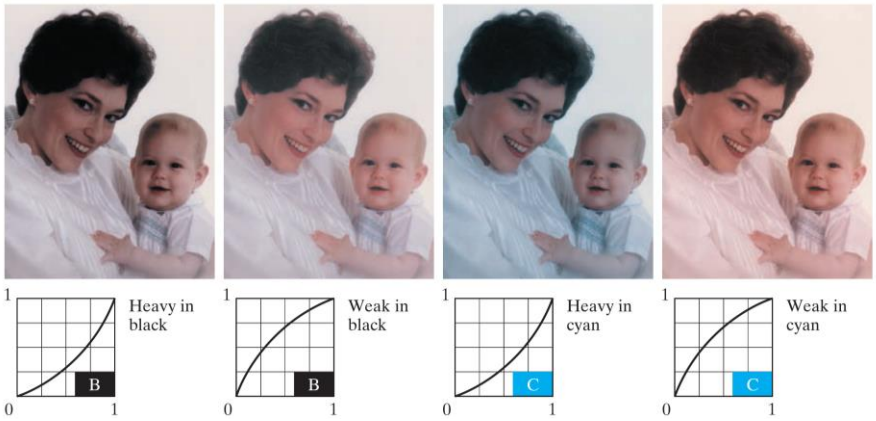


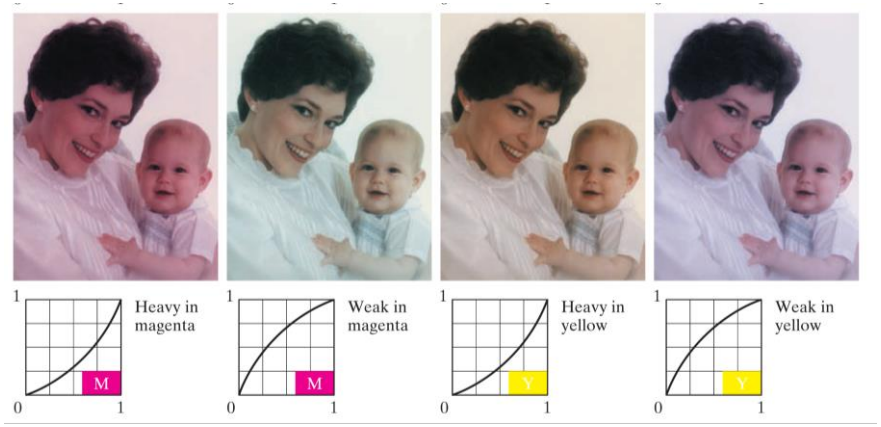



# CMYK



Original/Corrected



# Color Image Smoothing

The average of the RGB component vectors in this neighborhood:

$$\bar{\mathbf{c}}(x,y) = \frac{1}{K}\sum_{(s,t)\in S_{xy}} \mathbf{c}(s,t)$$

$$\bar{\mathbf{c}}(x,y) = \begin{bmatrix} \frac{1}{K}\sum_{(s,t)\in S_{xy}} R(s,t) \\ \frac{1}{K}\sum_{(s,t)\in S_{xy}} G(s,t) \\ \frac{1}{K}\sum_{(s,t)\in S_{xy}} B(s,t) \end{bmatrix}$$

# Color Image Smoothing



RGB image



Red component image



Green component



Blue component



(a) Hue. Intensity.



(b) Saturation.



(c)

# Color Image Smoothing



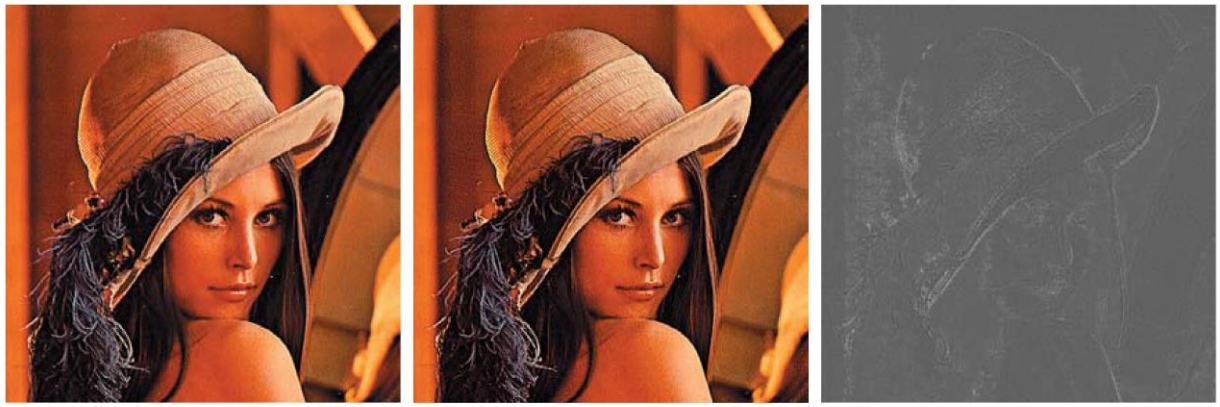
a b c

**FIGURE 6.40** Image smoothing with a $5 \times 5$ averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

# Color Image Smoothing

RGB color system, the Laplacian of vector **c**:

$$\nabla^2[\mathbf{c}(x, y)] = \begin{bmatrix} \nabla^2 R(x, y) \\ \nabla^2 G(x, y) \\ \nabla^2 B(x, y) \end{bmatrix}$$



a b c

**FIGURE 6.41** Image sharpening with the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.

# Thank You